Class 13

The indifference principle

Suppose σ^* is an NE. For player i, there are two strategies s_i^1, s_i^2 s.t.

$$\sigma_i^*(s_i^1) > 0, \quad \sigma_i^*(s_i^2) > 0$$

$$\Rightarrow U_i(s_i^1, \sigma_{-i}^*) = U_i(s_i^2, \sigma_{-i}^*)$$

Proof Suppose $U_i(s_i^1, \sigma_{-i}^*) > U_i(s_i^2, \sigma_{-i}^*)$

then
$$\bar{\sigma}_i(t_i) = \begin{cases} \sigma_i^*(t_i) & \text{if } t_i \neq s_i^1, s_i^2 \\ 0 & \text{if } t_i = s_i^2 \\ \sigma_i^*(s_i^1) + \sigma_i^*(s_i^2) & \text{if } t_i = s_i^1 \end{cases}$$

$$\boxed{\begin{aligned} &U_i(\sigma_i^*, \sigma_{-i}^*) \\ &= \sigma_i^*(s_i^1)\, U_i(s_i^1, \sigma_{-i}^*) \\ &\quad + (1 - \sigma_i^*(s_i^1)) \\ &\qquad U_i(s_i^2, \sigma_{-i}^*) \\ &U_i(s_i^1, \sigma_{-i}^*) \\ &\quad > U_i(s_i^2, \sigma_{-i}^*) \end{aligned}}$$

We can show that

$$U_i(\sigma_i^*, \sigma_{-i}^*) < U_i(\bar{\sigma}_i, \sigma_{-i}^*).$$

Corollary For an NE σ^*

$$U_i(\sigma^*) = U_i(s_i, \sigma_{-i}^*) \quad \forall s_i \in S_i \text{ s.t. } \sigma_i^*(s_i) > 0$$

Note: (s_i, σ_{-i}^*) may not be an NE.

- σ^* is an NE $\sigma_i^*(s_i) = 0$

$$U_i(\sigma^*) \geq U_i(s_i, \sigma_{-i}^*)$$

- σ^* is an NE, $U_i(\sigma^*) = U_i(s_i, \sigma_{-i}^*)$

for some $s_i \in S_i$

$$\not\Rightarrow \sigma_i^*(s_i) > 0$$

Example

	L	R
T	(4,4)	(,3)
B	(4,)	( )

Example

	1\2	y L	$1-y$ R
x	T	(1,-1)	(0,2)
$1-x$	B	(0,1)	(2,0)

T

$x(L) > 0$ $\quad 1-x(R) > 0$

$U_2(T,L) = U_2(T,R)$

- No pure strategy NE
- No NE where one player plays a pure strategy and the other player plays a proper mixed strategy
- $0<x<1$, $0<y<1$ $\quad U_1(T,y) = U_1(B,y)$

$$\Rightarrow y + 0(1-y) = 0 \cdot y + 2(1-y)$$

$$\Rightarrow y = 2(1-y)$$

$$\Rightarrow y = 2/3$$

$$U_2(x, L) = U_2(x, R)$$

$$\Rightarrow -x + (1-x) = 2x$$

$$\Rightarrow x = \frac{1}{4}$$

Finding NE of two-player games

$S_1 \quad S_2$

Fix $Y_1 \subseteq S_1$, $Y_2 \subseteq S_2$

$$\text{Supp}(\sigma_1) = \{s_1 \in S_1 \mid \sigma_1(s_1) > 0\}$$

$$\text{Supp}(\sigma_2) = \{s_2 \in S_2 \mid \sigma_2(s_2) > 0\}$$

Our goal is to check if there is an NE (σ_1^*, σ_2^*) s.t. $\text{Supp}(\sigma_i^*) = Y_i \quad \forall i = 1, 2.$

We have to find the solutions of the following linear equations

$|S_1| = m_1$

$|S_2| = m_2$

$(2^{m_1} - 1) \times (2^{m_2} - 1)$

1) $$\sum_{s_2 \in S_2} \sigma_2^*(s_2)\, u_1(s_1^0, s_2) \quad \boxed{U_1(s_1^0, \sigma_2^*) = U_1(s_1, \sigma_2^*)}$$

$$= \sum_{s_2 \in S_2} \sigma_2^*(s_2)\, u_1(s_1, s_2) \quad \forall s_1 \in Y_1$$

where $s_1^0 \in Y_1$

2) $$\sum_{s_1 \in S_1} \sigma_1^*(s_1)\, u_2(s_1, s_2^0) = \sum_{s_1 \in S_1} \sigma_1^*(s_1)\, u_2(s_1, s_2)$$

$\forall s_2 \in Y_2$ where

$s_2^0 \in Y_2$

3) $$\sum_{s_2 \in S_2} \sigma_1^0(s_2)\, u_1(s_1^0, s_2) \geq \sum_{s_2 \in S_2} \sigma_1^0(s_2)\, u_1(s_1, s_2) \quad \forall s_1 \notin Y_1$$

4) $$\sum_{s_1 \in S_1} \sigma_1^0(s_1)\, u_2(s_1, s_2^0) \geq \sum_{s_1 \in S_1} \sigma_1^0(s_1)\, u_2(s_1, s_2) \quad \forall s_2 \notin Y_2$$

5) $$\sum_{s_1 \in S_1} \sigma_1^0(s_1) = 1 = \sum_{s_2 \in S_2} \sigma_2^0(s_2)$$

6) $$\sigma_1^0(s_1) > 0, \quad \sigma_2^0(s_2) > 0 \quad \forall (s_1, s_2) \in (Y_1 \times Y_2)$$

7) $$\sigma_1^0(s_1) = \sigma_2^0(s_2) = 0 \quad \forall s_1 \notin Y_1, \ \forall s_2 \notin Y_2$$

Exercise: Show that σ_1^0 and σ_2^0 satisfying (1)-(7) is an NE.

- For general n-player games.

Fix $Y_1, \dots, Y_n$ of $S_1, \dots, S_n$

$$(2^{m_1}-1) \times (2^{m_2}-1) \times \cdots (2^{m_2}-1).$$

Example

1 \ 2	L	C	R
T	6,2	0,6	4,4
M	2,12	4,3	2,5
B	0,6	10,0	2,2

$\left(\frac{1}{2}(T), \frac{1}{2}(B)\right)$ strictly dominates M